

Simulation and Monte Carlo: Project 3

Likelihood-free inference and sums of log-normal variates

Questions 1 and 2

1. Theory of Likelihood-Free Inference: Reject-ABC

1.1 The Reject-ABC Algorithm and Point Estimation

As introduced in the course (slides p. 155), Approximate Bayesian Computation (ABC) is employed when the likelihood function $L(\theta) = p(\mathbf{y}^*|\theta)$ is numerically intractable but simulation from the generative model $p(\cdot|\theta)$ is feasible.

For a given observation y_i , the generative process is the sum of L independent log-normal random variables (as described in Bernton et al., 2019, Section 4.2):

$$y_i = \sum_{l=1}^L \exp(X_{i,l}), \quad X_{i,l} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2) \quad (1)$$

where $\theta = (\mu, \sigma^2)$. Because this sum has no closed-form density, we use the Reject-ABC algorithm, which targets the joint distribution:

$$p_\epsilon(\theta, \mathbf{y}|\mathbf{y}^*) = \frac{\pi(\theta)p(\mathbf{y}|\theta)\mathbb{I}(d(s(\mathbf{y}), s(\mathbf{y}^*)) \leq \epsilon)}{\mathbb{P}(d(s(\mathbf{y}), s(\mathbf{y}^*)) \leq \epsilon)} \quad (2)$$

where $\mathbb{I}(\cdot)$ is the indicator function, $s(\cdot)$ is a summary statistic, and $d(\cdot, \cdot)$ is a distance metric. Following the article (Section 3.1.1), we use the 1-Wasserstein distance W_1 computed via sorted order statistics: $W_1(\mathbf{y}, \mathbf{y}^*) = \frac{1}{n} \sum_{i=1}^n |y_{(i)} - y_{(i)}^*|$.

To estimate θ , we consider the ABC posterior mean, which is the expectation of θ under the marginal ABC posterior $\pi_\epsilon(\theta|\mathbf{y}^*)$:

$$\hat{\theta}_{ABC} = \mathbb{E}_{\pi_\epsilon}[\theta] = \int_{\Theta} \theta \pi_\epsilon(\theta|\mathbf{y}^*) d\theta = \frac{\int_{\Theta} \theta \pi(\theta) L_\epsilon(\theta) d\theta}{\int_{\Theta} \pi(\theta) L_\epsilon(\theta) d\theta} \quad (3)$$

where $L_\epsilon(\theta) = \mathbb{P}(W_1(\mathbf{y}, \mathbf{y}^*) \leq \epsilon|\theta)$ acts as a "smoothed" or "approximate" likelihood.

1.2 The Role of the Tolerance ϵ

The parameter ϵ controls the fundamental trade-off between approximation bias and computational variance.

Theoretical Convergence ($\epsilon \rightarrow 0$): If $s(\mathbf{y})$ is a sufficient statistic, the course (slides p. 156) and the article (Section 1.3) state that $\pi_\epsilon(\theta|\mathbf{y}^*) \xrightarrow{\epsilon \rightarrow 0} \pi(\theta|\mathbf{y}^*)$. Mathematically, as ϵ shrinks, the acceptance region concentrates around the exact observation $\mathbf{y}^*$. In the limit, the indicator function behaves as a Dirac measure $\delta_{s(\mathbf{y}^*)}(s(\mathbf{y}))$, recovering the exact posterior.

Degeneration to Prior ($\epsilon \rightarrow \infty$): Conversely, when ϵ is large enough to encompass the entire support of $p(\mathbf{y}|\theta)$, the indicator function $\mathbb{I}(W_1(\mathbf{y}, \mathbf{y}^*) \leq \epsilon)$ is identically 1. Thus, $L_\epsilon(\theta) = \int p(\mathbf{y}|\theta) d\mathbf{y} = 1$. The ABC posterior then simplifies to:

$$\pi_\infty(\theta|\mathbf{y}^*) = \frac{\pi(\theta) \cdot 1}{\int \pi(\theta') d\theta'} = \pi(\theta) \quad (4)$$

The inference yields no information beyond the prior beliefs.

1.3 The Influence of the Prior $\pi(\theta)$ and hyperparameters (a, b, κ)

Following the project guidelines, we utilize a fully conjugate Normal-Inverse-Gamma prior structure: $\sigma^2 \sim \text{IG}(a, b)$ with $a, b > 0$, and $\mu | \sigma^2 \sim \mathcal{N}(0, \kappa \sigma^2)$. The choice of the hyperparameters a, b , and κ directly governs the estimation dynamics.

Prior Bias (Concentrated Prior): Unlike standard MCMC where the prior’s influence vanishes with large n (Bernstein-von Mises theorem), in ABC, the prior directly shapes the proposal distribution. If the prior is overly concentrated (e.g., a very small κ tightly bounding μ around 0, or a large a with a specific b creating a narrow peak for σ^2) in a region of low likelihood, the ABC posterior will be severely biased towards the prior mean. The algorithm struggles to draw samples in the true high-probability region, forcing the inference to rely solely on the narrow prior bounds.

Algorithmic Efficiency (Diffuse Prior): The efficiency is measured by the overall acceptance probability $\alpha = \int \pi(\theta) L_\epsilon(\theta) d\theta$. If the prior is highly diffuse (large κ spreading the conditional mean, and small a, b flattening the variance distribution), the probability mass $\pi(\theta)$ is spread out thinly over Θ . Since the approximate likelihood $L_\epsilon(\theta)$ is only significant in a small neighborhood of the true parameters, the integral α becomes vanishingly small. This leads to a severe drop in the acceptance rate, causing extreme computational inefficiency as the vast majority of simulated datasets are rejected.

1.4 Empirical Results and Interpretation

To empirically validate the theoretical effects of the prior hyperparameters a, b , and κ , the Rejection-ABC algorithm was executed on synthetic data generated with true parameters $\mu = 0.0$, $\sigma = 0.3$, $L = 10$, and $n = 200$. The tolerance was fixed at $\epsilon = 0.5$, and a computational ceiling of 10^4 iterations was enforced. A strict random number generator (seed = 42) was utilized to ensure exact reproducibility.

- **Well-calibrated Prior** ($a = 3.0, b = 0.2, \kappa = 5.0$): With these parameters, the Inverse-Gamma prior is centered reasonably close to the true variance ($\mathbb{E}[\sigma^2] = \frac{b}{a-1} = 0.1$), and the conditional Normal allows sufficient exploration for μ . The algorithm successfully approximated the parameters, yielding posterior means of $\hat{\mu} \approx -0.0053$ and $\hat{\sigma} \approx 0.2620$ (target: 0.0 and 0.3). The acceptance rate was stable at approximately 4.97%.
- **Highly Diffuse Prior** ($a = 0.1, b = 0.1, \kappa = 100.0$): As predicted, inflating the prior variance severely degrades algorithmic efficiency. The acceptance rate collapsed to 0.03%, yielding only 3 accepted samples before reaching the 10^4 iteration limit. Furthermore, this configuration triggered numerical overflows (`RuntimeWarning: overflow encountered in exp`). The extremely diffuse prior proposes absurdly large values for μ and σ^2 , generating synthetic latent variables X so large that their exponentiation exceeds standard floating-point limits. This perfectly illustrates how uninformative priors can lead to computational failure in generative models involving exponentials.
- **Highly Concentrated Prior** ($a = 100.0, b = 100.0, \kappa = 0.01$): This configuration demonstrates extreme prior bias. With $a = 100.0$ and $b = 100.0$, the Inverse-Gamma distribution is tightly peaked around $\sigma^2 \approx 1$ (i.e., $\sigma \approx 1$). Given that the true data-generating scale is $\sigma = 0.3$, the simulated datasets consistently exhibit excessive variance relative to the observed data. Furthermore, $\kappa = 0.01$ heavily restricts μ near 0. Consequently, the Wasserstein distance never satisfies the condition $W_1 \leq 0.5$. The acceptance rate drops precisely to 0% (0 samples accepted), resulting in a complete failure of the inference process due to the severe misspecification of the highly informative prior.

These empirical findings strictly corroborate the theoretical property that ABC inference is uniquely sensitive to prior specification, which governs both the computational viability of the sampler and the asymptotic bias of the resulting posterior estimate.

2. The MCMC-ABC Algorithm

2.1 Derivation of the Sampler and Reparameterization

The fundamental limitation of Rejection-ABC, as demonstrated empirically in Question 1, is that proposals are drawn blindly from the prior $\pi(\theta)$. If the prior is diffuse, the acceptance probability collapses.

To mitigate this, we construct a Markov Chain Monte Carlo (MCMC) sampler that targets the identical approximate posterior, following Marjoram et al. (2003).

Let the target invariant distribution be the joint ABC posterior on the extended state space $(\Theta \times \mathcal{Y})$. As defined in the course, this target is:

$$\pi_\epsilon(\theta, \mathbf{z}|\mathbf{y}^*) = \frac{\pi(\theta)p(\mathbf{z}|\theta)\mathbb{I}(W_1(\mathbf{z}, \mathbf{y}^*) \leq \epsilon)}{Z_\epsilon} \quad (5)$$

To construct our Metropolis-Hastings (MH) algorithm, we must specify a proposal distribution. The parameter σ^2 is strictly positive. A standard Gaussian random walk on σ^2 could propose illegal negative values. To prevent this, we reparameterize the problem and perform the random walk on the unbounded space of $\log(\sigma^2)$. Our proposed state $(\mu', \log(\sigma'^2))$ is generated via a symmetric Gaussian random walk:

$$\mu' = \mu + \eta_\mu, \quad \eta_\mu \sim \mathcal{N}(0, \text{step}_\mu^2) \quad (6)$$

$$\log(\sigma'^2) = \log(\sigma^2) + \eta_\sigma, \quad \eta_\sigma \sim \mathcal{N}(0, \text{step}_\sigma^2) \quad (7)$$

Given the proposed parameters θ' , we simulate a new dataset $\mathbf{z}' \sim p(\cdot|\theta')$.

Because the random walk is symmetric in the proposed space, the Hastings proposal ratio is 1. However, because our target prior $\pi(\theta)$ is defined in terms of (μ, σ^2) , we must include the Jacobian of the transformation $\sigma^2 = \exp(\log(\sigma^2))$ when evaluating the acceptance ratio. The Jacobian is precisely σ^2 .

The probability α of accepting the proposed state simplifies mathematically to:

$$\alpha = \min \left(1, \frac{\pi(\mu', \sigma'^2)}{\pi(\mu, \sigma^2)} \times \frac{\sigma'^2}{\sigma^2} \times \mathbb{I}(W_1(\mathbf{z}', \mathbf{y}^*) \leq \epsilon) \right) \quad (8)$$

In our implementation, to prevent numerical underflows, we compute this ratio entirely in the log-domain. The log-acceptance ratio becomes:

$$\log(\alpha) = \Delta \log \pi(\mu|\sigma^2) + \Delta \log \pi(\sigma^2) + \Delta \log(\sigma^2) \quad (9)$$

where $\Delta \log(\sigma^2) = \log(\sigma'^2) - \log(\sigma^2)$ acts as our log-Jacobian adjustment. If the simulated data satisfies the Wasserstein distance condition $(W_1 \leq \epsilon)$ and $U \sim \mathcal{U}(0, 1) \leq \alpha$, the proposed state is accepted.

2.2 Proof of Detailed Balance

To prove that this MCMC algorithm converges to the correct target distribution π_ϵ , we rely on the fundamental theorem of MCMC convergence: we must demonstrate that the transition kernel satisfies the detailed balance (reversibility) condition.

Let $\mathcal{K}(x'|x)$ be the total transition kernel of moving from state $x = (\theta, \mathbf{z})$ to $x' = (\theta', \mathbf{z}')$. By definition of the MH algorithm:

$$\mathcal{K}(x'|x) = Q(x'|x)\alpha(x \rightarrow x') + \delta_x(x') \left(1 - \int Q(x''|x)\alpha(x \rightarrow x'')dx'' \right) \quad (10)$$

We evaluate the reversibility condition for $x \neq x'$:

$$\pi_\epsilon(x|\mathbf{y}^*)\mathcal{K}(x'|x) = \pi_\epsilon(x|\mathbf{y}^*)Q(x'|x) \min \left(1, \frac{\pi_\epsilon(x'|\mathbf{y}^*)Q(x|x')}{\pi_\epsilon(x|\mathbf{y}^*)Q(x'|x)} \right) \quad (11)$$

$$= \min \left(\pi_\epsilon(x|\mathbf{y}^*)Q(x'|x), \pi_\epsilon(x'|\mathbf{y}^*)Q(x|x') \right) \quad (12)$$

Because the minimum function is symmetric with respect to its arguments, we can perfectly reverse the roles of x and x' :

$$\min \left(\pi_\epsilon(x|\mathbf{y}^*)Q(x'|x), \pi_\epsilon(x'|\mathbf{y}^*)Q(x|x') \right) = \pi_\epsilon(x'|\mathbf{y}^*)Q(x|x') \min \left(\frac{\pi_\epsilon(x|\mathbf{y}^*)Q(x'|x)}{\pi_\epsilon(x'|\mathbf{y}^*)Q(x|x')}, 1 \right) \quad (13)$$

Which equates to $\pi_\epsilon(x'|\mathbf{y}^*)\mathcal{K}(x|x')$. Thus, the detailed balance equation holds. The Markov chain is reversible with respect to π_ϵ , ensuring that it targets exactly the same approximate posterior as Rejection-ABC.

2.3 Algorithmic Superiority (Discussion)

The MCMC-ABC sampler fundamentally alters the search dynamics, making it superior to Rejection-ABC in specific, problematic regimes.

Resilience to Diffuse Priors: In Rejection-ABC, the proposal distribution is the prior. As shown in Question 1, if the prior is highly diffuse (e.g., extremely large κ and small a, b), the probability mass spreads thinly over the parameter space. The algorithm blindly proposes extreme values, leading to near-zero acceptance rates and severe numerical overflows. MCMC-ABC solves this by using a local proposal $q(\theta'|\theta)$. Once the Markov chain locates a region of high approximate likelihood, it explores locally around that region. The efficiency becomes driven by the variance of the random walk q , and is asymptotically uncoupled from the global dispersion of the prior $\pi(\theta)$. MCMC-ABC can therefore perform efficiently even with highly uninformative priors.

Handling Small Tolerances (ϵ): To reduce the theoretical approximation bias (as $\epsilon \rightarrow 0$), the volume of the acceptance region shrinks drastically. Rejection-ABC must blindly draw from the entire prior space hoping to land in this microscopic volume. MCMC-ABC, however, proposes small perturbations from a state that is *already* accepted (inside the ϵ -ball). The probability of a local perturbation remaining within the ϵ -ball is exponentially higher than a random draw from the global space, allowing MCMC-ABC to tolerate significantly smaller values of ϵ while maintaining a viable computational variance.

3. Exact MCMC Sampler via Data Augmentation

3.1 Derivation of the Joint Target Distribution

To bypass the intractable sum of log-normal variates, we adopt a data augmentation strategy. By treating the unobserved Gaussian random variables $X_{i,l}$ as latent variables, the extended state space becomes $(\theta, \mathbf{X})$, where $\mathbf{X} = \{X_{i,l}\}$. The joint target distribution is proportional to the product of the prior, the likelihood of the latent variables, and a deterministic constraint linking the latent variables to the observations $\mathbf{y} = \{y_i\}_{i=1}^n$:

$$p(\theta, \mathbf{X} | \mathbf{y}) \propto \pi(\theta) \left(\prod_{i=1}^n \prod_{l=1}^L \mathcal{N}(X_{i,l} | \mu, \sigma^2) \right) \prod_{i=1}^n \delta \left(y_i - \sum_{l=1}^L \exp(X_{i,l}) \right) \quad (1)$$

For mathematical and computational tractability (to ensure closed-form conditional distributions), we select conditionally conjugate priors: $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(\alpha_0, \beta_0)$.

3.2 Step (a): Gibbs Update for Parameters θ

Conditional on the latent variables $\mathbf{X}$, the deterministic observation constraints are automatically satisfied. The problem reduces to finding the posterior of $\theta = (\mu, \sigma^2)$ given $N = n \times L$ independent Gaussian observations.

Thanks to conjugacy, the conditional distributions perfectly factorize into standard families:

- **Mean μ :** $\mu | \mathbf{X}, \sigma^2 \sim \mathcal{N}(\mu_n, \sigma_n^2)$

$$\text{where } \sigma_n^2 = \left(\frac{1}{\sigma_0^2} + \frac{N}{\sigma^2} \right)^{-1} \quad \text{and} \quad \mu_n = \sigma_n^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{\sum_{i,l} X_{i,l}}{\sigma^2} \right) \quad (2)$$

- **Variance σ^2 :** $\sigma^2 | \mathbf{X}, \mu \sim \text{Inverse-Gamma}(\alpha_n, \beta_n)$

$$\text{where } \alpha_n = \alpha_0 + \frac{N}{2} \quad \text{and} \quad \beta_n = \beta_0 + \frac{1}{2} \sum_{i,l} (X_{i,l} - \mu)^2 \quad (3)$$

3.3 Step (b): Metropolis Updates for Latent Variable Pairs

We cannot update a single $X_{i,l}$ using a standard Gibbs or Metropolis step, as any unilateral perturbation would violate the deterministic constraint $y_i = \sum_{l=1}^L \exp(X_{i,l})$. Consequently, we update blocks of two variables, $(X_{i,l}, X_{i,l'})$, such that their partial sum $S = \exp(X_{i,l}) + \exp(X_{i,l'})$ remains strictly invariant.

We parameterize the sub-manifold of valid configurations using the log-difference $w = X_{i,l} - X_{i,l'}$. The inverse mapping from the fixed S and the free variable w back to the original variables is

uniquely determined:

$$X_{i,l} = \log S + w - \log(1 + e^w) \quad (4)$$

$$X_{i,l'} = \log S - \log(1 + e^w) \quad (5)$$

The Jacobian determinant of the transformation $(X_{i,l}, X_{i,l'}) \rightarrow (S, w)$ is exactly $1/S$. Because $1/S$ is constant with respect to w , the conditional target density for w given S simplifies elegantly to the product of the priors:

$$p(w|S, \mu, \sigma^2) \propto \exp\left(-\frac{(X_{i,l}(w) - \mu)^2}{2\sigma^2} - \frac{(X_{i,l'}(w) - \mu)^2}{2\sigma^2}\right) \quad (6)$$

By proposing a symmetric random walk $w^* \sim \mathcal{N}(w, \tau^2)$, the Metropolis-Hastings acceptance probability avoids intractable Jacobian terms and simplifies to:

$$\alpha = \min\left(1, \frac{p(w^*|S, \mu, \sigma^2)}{p(w|S, \mu, \sigma^2)}\right) \quad (7)$$

During Step (b), we randomly sample multiple distinct pairs (l, l') for each observation i and apply this computationally lightweight Metropolis step to rapidly mix the latent configuration.

4. Empirical Assessment of ABC Bias

To rigorously assess the bias introduced by the ABC approximation, we implemented the Exact Gibbs Sampler derived in Question 3 and benchmarked it against Reject-ABC. The synthetic data replicates the setup in Question 1: $n = 200$, $L = 10$, with true parameters $\mu = 0.0$ and $\sigma = 0.3$.

Both samplers leveraged identical conjugate priors: $\mu \sim \mathcal{N}(0, 1)$ and $\sigma^2 \sim \text{Inverse-Gamma}(2.0, 0.2)$. The Exact MCMC posterior converged tightly around the true parameters. For Rejection-ABC, we simulated 5×10^4 candidates, using the 1-Wasserstein distance between sorted observations as the summary statistic.

Bias Evaluation Table:

Method	Tolerance (ϵ)	Mean μ	Bias μ	Mean σ	Bias σ
Exact MCMC	Exact (0.0)	0.0136	0.0000	0.2967	0.0000
Reject-ABC	$\epsilon = 2.00$	-0.0347	-0.0483	0.3677	0.0710
Reject-ABC	$\epsilon = 1.00$	-0.0071	-0.0208	0.3398	0.0431
Reject-ABC	$\epsilon = 0.50$	0.0066	-0.0070	0.3074	0.0106
Reject-ABC	$\epsilon = 0.25$	0.0114	-0.0023	0.2957	-0.0011

Interpretation: The exact Gibbs sampler completely eliminates the approximation gap, delivering an unbiased posterior (relative to the theoretical Bayesian posterior). The ABC evaluation distinctly highlights the bias-variance trade-off parameterized by ϵ :

- **High Tolerance ($\epsilon = 2.0$):** The ABC acceptance region is overly permissive, causing the empirical posterior to heavily revert toward the prior distributions. This induces massive bias, particularly artificially inflating the variance parameter.
- **Low Tolerance ($\epsilon \leq 0.25$):** The bias shrinks significantly as the approximation distribution narrowly concentrates around the true exact posterior. However, this statistical accuracy comes at the cost of crippling algorithmic inefficiency, as the acceptance rate plummets to near-zero.